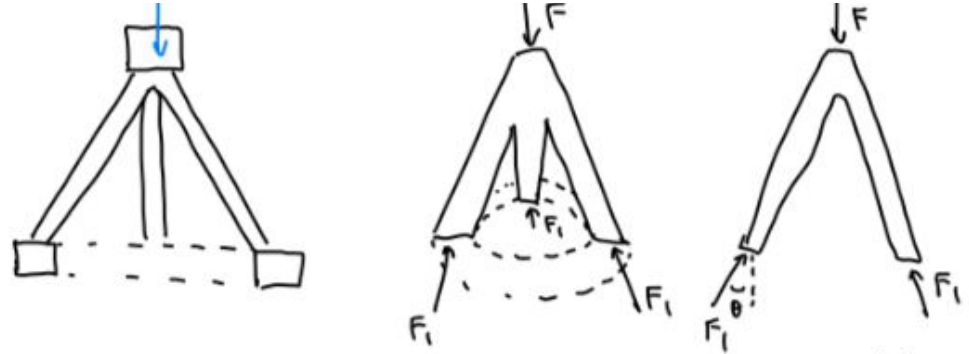


Free Body Diagrams

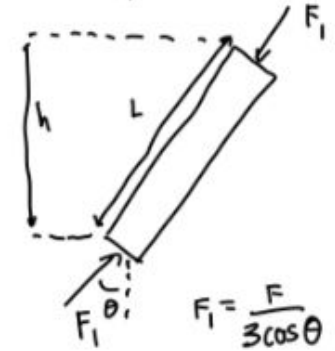


- Axial force distribution depends on the angle of the legs

Compression :

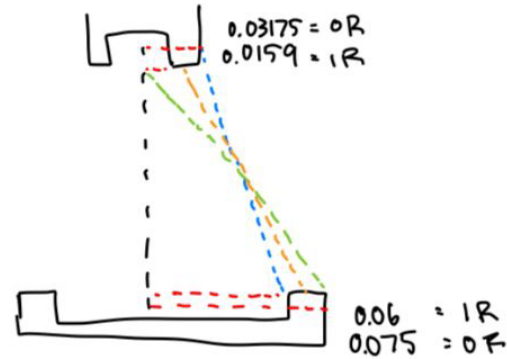
$$\text{Stress} = \sigma = \frac{F}{A}$$

$$\text{failure} \rightarrow \text{buckling} = \frac{C\pi^2 EI}{L^2}$$

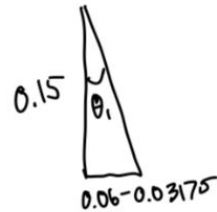


$$\sigma = \frac{F_1}{A} = \frac{F}{3 \cos \theta \cdot A} = \frac{S_y}{FOS}$$

Free Body Diagrams



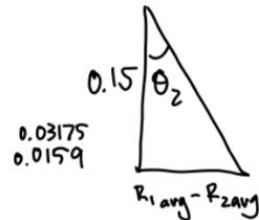
steepest possible:



$$\theta_1 = \tan^{-1} \left(\frac{0.06 - 0.03175}{0.15} \right)$$

$$\theta_1 = 10.67^\circ$$

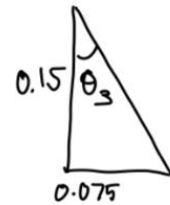
mid



$$\theta_2 = \tan^{-1} \left(\frac{\frac{0.06 + 0.075}{2} - \frac{0.03175 + 0.0159}{2}}{0.15} \right)$$

$$\theta_2 = 16.23^\circ$$

shallowest possible



$$\theta_3 = \tan^{-1} \left(\frac{0.075}{0.15} \right)$$

$$\theta_3 = 26.57^\circ$$

Inverse Failure Analysis

Axial Stress Analysis

$$\sigma = \frac{F_1}{A} = \frac{F}{3 \cos \theta \cdot (\pi r^2)} = \frac{S_y}{FoS}$$

$$r^2 = \frac{F \cdot FoS}{3 \cos \theta \cdot \pi \cdot S_y}$$

$$r = \sqrt{\frac{F \cdot FoS}{3 \cos \theta \cdot \pi \cdot S_y}}$$

Buckling Stress Analysis

$$F_{cr} = F_1 \cdot FoS = \frac{C \pi^2 E I}{L^2}$$

$$C = 1 = \text{Pinned / pinned}$$

$$I = \frac{1}{4} \pi r^4$$

$$\frac{F \cdot FoS}{n \cos \theta} = \frac{C \pi^2 E \frac{1}{4} \pi r^4}{L^2}$$

$$\frac{F \cdot FoS}{n \cos \theta} = \frac{C \pi^2 E \cdot \pi r^4 \cos^2 \theta}{4 h^2}$$

$$r^4 = \frac{F \cdot FoS \cdot 4 h^2}{n \cos \theta \cdot \pi^2 \cdot E \cdot \pi \cdot \cos^2 \theta} = \frac{F \cdot FoS \cdot 4 h^2}{C n \cos^3 \theta \cdot \pi^3 E}$$

x-section:



Inverse Failure Analysis

Axial Stress Analysis

$$r = 6.65 \cdot 10^{-4} \text{ m} = 0.665 \text{ mm} \rightarrow \text{FoS} = 1$$

$$r = 8.14 \cdot 10^{-4} \text{ m} = 0.814 \text{ mm} \rightarrow \text{FoS} = 1.5$$

$$r = 9.40 \cdot 10^{-4} \text{ m} = 0.940 \text{ mm} \rightarrow \text{FoS} = 2$$

Yield Strength: 50 MPa

Elastic Modulus: 3.6 GPa

Density: $1.24 \text{ g} \cdot \text{cm}^{-3}$

$$\theta = 16.23^\circ$$

$$C = 2$$

Buckling Stress Analysis

$$r = \sqrt[4]{\frac{200\text{N} \cdot 1 \cdot 4 \cdot (0.15)^2 \cdot 1}{2 \cdot 3 \cdot \cos^3(16.23^\circ) \pi^3 \cdot 3.6 \cdot 10^9}}$$

$$\text{FoS} = 1 = 0.00235$$

$$r = \sqrt[4]{\frac{200\text{N} \cdot 1.5 \cdot 4 \cdot (0.15)^2 \cdot 1.5}{2 \cdot 3 \cdot \cos^3(16.23^\circ) \pi^3 \cdot 3.6 \cdot 10^9}}$$

$$\text{FoS} = 1.5 = 0.002598$$

$$r = \sqrt[4]{\frac{200\text{N} \cdot 2 \cdot 4 \cdot (0.15)^2 \cdot 2}{2 \cdot 3 \cdot \cos^3(16.23^\circ) \pi^3 \cdot 3.6 \cdot 10^9}}$$

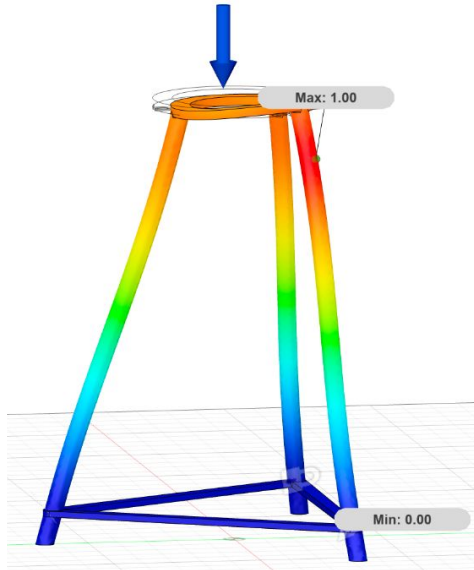
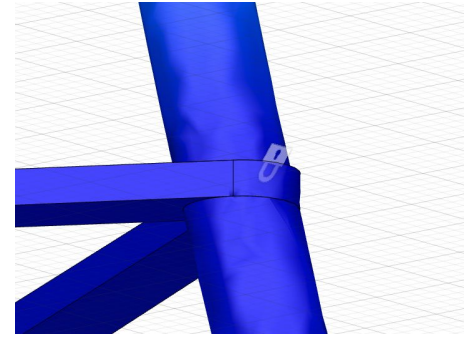
$$\text{FoS} = 2 = 0.002792$$

FEA Results – Buckling

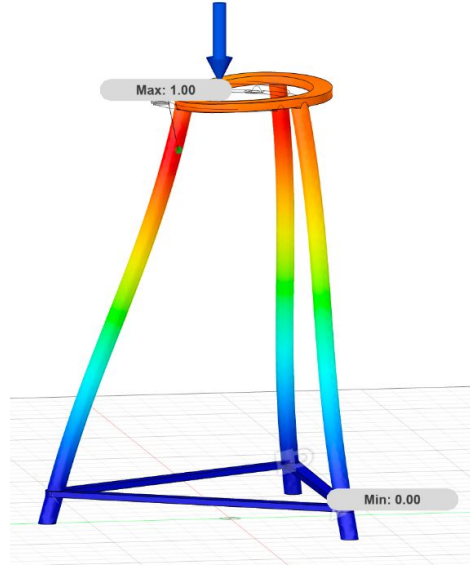
Buckling Mode 1: 1.769 x Load

Buckling Mode 2: 1.771 x Load

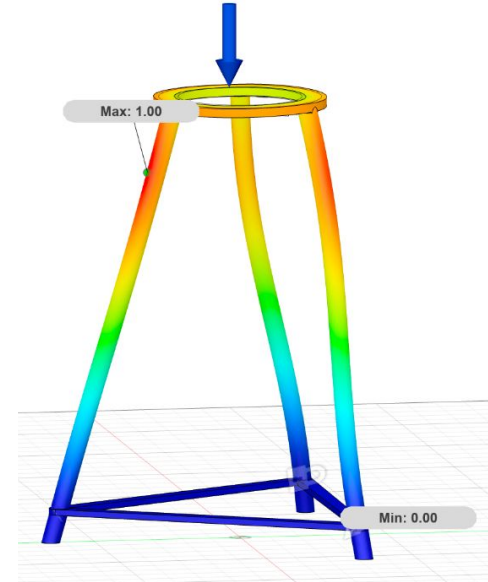
Buckling Mode 3: 1.904 x Load



Buckling Mode 1



Buckling Mode 2

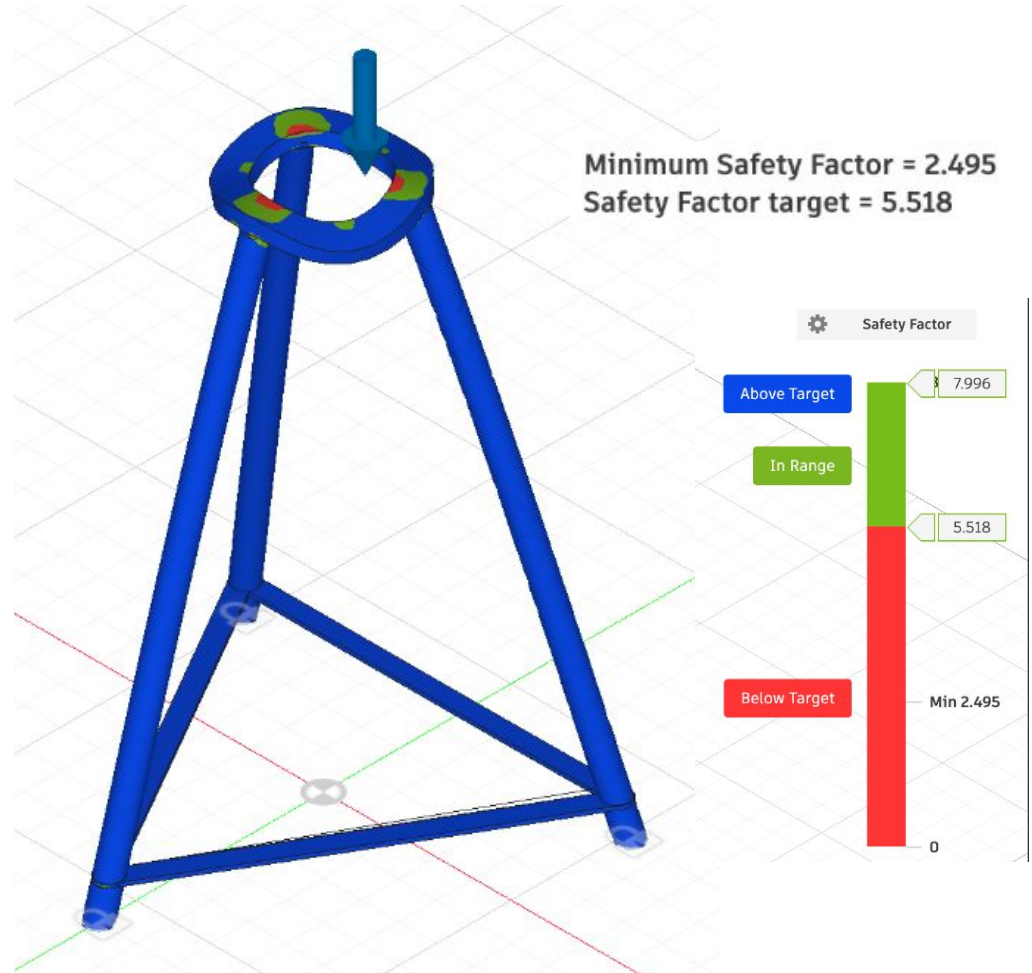


Buckling Mode 3

FEA Results – Axial

$$\delta = \frac{F_i L}{AE} = \frac{F}{3 \cos \theta} \cdot \frac{h}{\cos \theta} = \frac{200 \text{ N}}{3 \cos(16.23^\circ)} \cdot \frac{0.15}{\cos(16.23^\circ)} = \frac{200 \text{ N}}{\pi (0.002742)^2 \cdot 36 \cdot 10^9}$$

$$\delta \approx 0.1 \text{ mm}$$



Summary

- 3 legged tripod at a 16.23° with leg radius of 2.792 mm
- Failure due to buckling
- Factor of Safety: 2
- Final weight: 14 grams

